



SAINT LOUIS UNIVERSITY

—
**SCHOOL OF SCIENCE
AND ENGINEERING**

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Artemis Mission Data Explorer

By

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Abstract

The Artemis Mission Data Explorer is an interactive MATLAB-based program designed to make information about NASA's Artemis missions more accessible to a general audience. Space exploration data is often presented in technical formats that are difficult for non-experts to understand. This project addresses that issue by organizing mission details into a structured, menu-driven interface that allows users to easily explore key aspects of the Artemis program. The application includes mission summaries, educational explanations, data visualizations, and links to external NASA resources. By combining simple navigation with graphical representations, the program helps users better understand mission timelines, crew sizes, and the overall purpose of lunar exploration. The full project, including source code and documentation, is available on [GitHub](https://github.com/Yasssiin/matlab-artemis-explorer) at: [https://github.com/Yasssiin/matlab-artemis-explorer]. This project demonstrates how programming and data organization can be used to simplify complex engineering topics and improve accessibility.

Introduction

Space exploration has always been an important part of scientific and technological advancement, and NASA's Artemis program represents the next major step in human exploration beyond Earth. The goal of Artemis is to return humans to the Moon and establish a sustainable presence, while also preparing for future missions to Mars. Despite its importance, much of the information about Artemis missions is scattered across different platforms and presented in a way that can be difficult for students and the general public to understand.

This project was developed to bridge that gap by creating a centralized, interactive tool that simplifies Artemis mission data. Using MATLAB, we designed a program that organizes mission details into an easy-to-use interface, allowing users to navigate through different sections without needing prior technical knowledge. The program focuses on clarity, accessibility, and engagement by combining structured data with simple visualizations and guided menu options.

The Artemis Mission Data Explorer not only provides factual information about missions such as Artemis I, II, and III, but also explains why lunar exploration is important and how each mission contributes to the overall program. By doing so, it serves as both an educational tool and an example of how engineering concepts can be communicated more effectively through interactive design.

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1. Problem Statement

Space exploration, specifically NASA's Artemis program, can be difficult for the general public to understand and access. Most available information is spread across multiple websites, written in highly technical language, and lacks interactivity. This makes it challenging for students and non-experts to quickly learn key mission details or understand how the missions connect to one another.

Our interest in this project comes from both a curiosity about space exploration and a desire to make complex engineering topics more accessible. We wanted to create a tool that simplifies Artemis mission information into something that is easy to navigate, visually engaging, and informative without requiring prior technical knowledge.

2. Project Goals and Functionality

The primary goal of this project was to develop an interactive MATLAB program that allows users to explore Artemis mission data in a simple and organized way.

The desired functionality includes:

- A **menu-driven interface** that allows users to navigate the program easily
- The ability to view **individual mission summaries** (Artemis I, II, and III)
- A section explaining **why lunar missions matter**
- A complete overview of the **entire Artemis program**
- A **timeline visualization** showing mission order and crew sizes
- A **trajectory visualization** of missions (Artemis I, II, and III).
- The ability to open **external NASA resources** (news and image gallery)

The program is designed to be intuitive so that users can select options without needing instructions or prior MATLAB experience.

3. Meeting Project Objectives

Overall, our project meets the objectives outlined in our proposal. We successfully created an interactive program that organizes Artemis mission data and presents it in a user-friendly format. All core features such as mission summaries, program explanation, and timeline visualization are fully functional.

However, there are a few limitations:

- The program currently includes only **three missions (Artemis I, II, and III)** and could be expanded in the future
- The interface is limited to MATLAB dialog boxes rather than a fully graphical user interface (GUI)
- The visualization is basic and could be enhanced with more advanced graphics

There are no major bugs in the code. One minor issue is that opening external websites depends on the user's system browser, which may occasionally fail. This is handled using error messages.

4. Technical Challenges and Solutions

One major technical challenge was **organizing a large amount of mission data** in a clean and manageable way. This was solved by using a structured array (`missionData`) that stores all mission properties such as name, launch year, crew size, and descriptions.

Another challenge was creating an **interactive user experience**. MATLAB is not primarily designed for user interfaces, so we implemented the built-in `menu()` function to simulate an interactive system. This allowed users to easily navigate between different sections.

A third challenge was **visualizing mission data**. We addressed this by creating a timeline using MATLAB plotting functions. A stem plot was used to represent mission years, and a bar chart was used to show crew sizes, making the data more engaging and easier to understand.

Finally, we needed to ensure that external links worked properly. This was handled using a `try/catch` structure to prevent the program from crashing if a browser fails to open.

5. Code Structure

Our program is primarily written as a **main function** (`artemis_mission_data_explorer`) with several supporting functions. We chose to use functions instead of a single script to improve organization, readability, and modularity.

The structure includes:

- A **main function** that controls program flow and user interaction
- A **data-building function** (`buildMissionData`) that initializes all mission information
- Several **display functions** that handle different sections of the program:
 - Mission summaries
 - Program explanations
 - Educational content
- A **visualization function** that generates graphs
- A **utility function** (`openWebsite`) for opening external links

Splitting the program into multiple functions allowed us to separate responsibilities and make the code easier to debug and expand.

6. Usage Instructions

To run the program:

1. Open MATLAB
2. Make sure the file `artemis_mission_data_explorer.m` is in your current directory
3. In the command window, type:
`artemis_mission_data_explorer`

4. Press Enter

Once the program runs, a menu will appear with several options:

- Select **Artemis I, II or III Overview** to view mission details
- Select **Why Lunar Missions Matter** for general information
- Select **Entire Artemis Program Explained** for a full overview
- Select **Mission Timeline Visualization** to view graphs
- Select **Artemis I, II or III Trajectory Overview** to view graphs for Orbit Trajectories
- Select **Open Artemis News** or **NASA Gallery** to open external links
- Select **Exit** to close the program

Inputs and Outputs

- **Input:** User selections from the menu
- **Output:** Dialog boxes displaying information and graphical plots

Function Purposes

- `artemis_mission_data_explorer()` – runs the program and menu
- `buildMissionData()` – stores mission data
- `showMissionSummary()` – displays mission details
- `showWhyLunarMissions()` – explains importance of lunar missions
- `showEntireArtemisExplanation()` – summarizes the program
- `showTimelineVisualization()` – generates graphs

- `openWebsite()` – opens external links

7. Conclusion

The Artemis Mission Data Explorer demonstrates how complex technical information can be transformed into a format that is both interactive and easy to understand. By using MATLAB as the development platform, this project successfully combines structured data storage, menu-driven navigation, and basic visualization techniques to create a cohesive learning tool. Users are able to explore mission details, understand the purpose of lunar exploration, and visualize important data such as mission timelines and crew sizes without needing prior background knowledge.

One of the key strengths of this project is its focus on accessibility. Instead of presenting raw data or technical documents, the program organizes information into clearly defined sections that guide the user through the Artemis program step by step. This makes it especially useful for students or individuals who are new to space exploration topics. Additionally, the inclusion of external NASA resources allows users to go beyond the program and explore real-world updates and imagery, further enhancing the educational value.

While the project meets its main objectives, there are several opportunities for future improvement. Expanding the dataset to include additional Artemis missions and related programs would provide a more complete picture of NASA's long-term goals. Improving the graphical interface, possibly by transitioning from simple dialog boxes to a full GUI, would enhance user experience and make the program more visually appealing. More advanced data visualizations, such as interactive plots or animations, could also make the information more engaging.

Overall, this project highlights the importance of presenting engineering and scientific information in a way that is understandable to a wider audience. The Artemis Mission Data Explorer serves as a strong example of how programming can be used not just for computation, but also for communication and education.

8. Figures

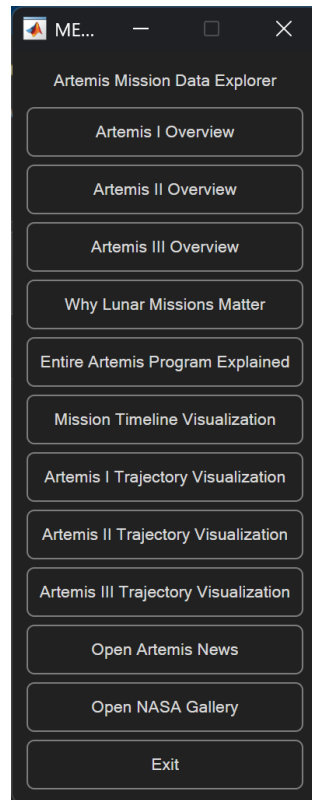


Figure (1): Menu Interface.

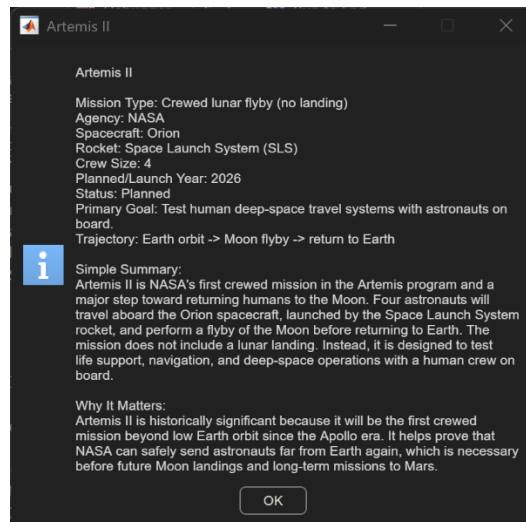


Figure (2): Example Information Interface.

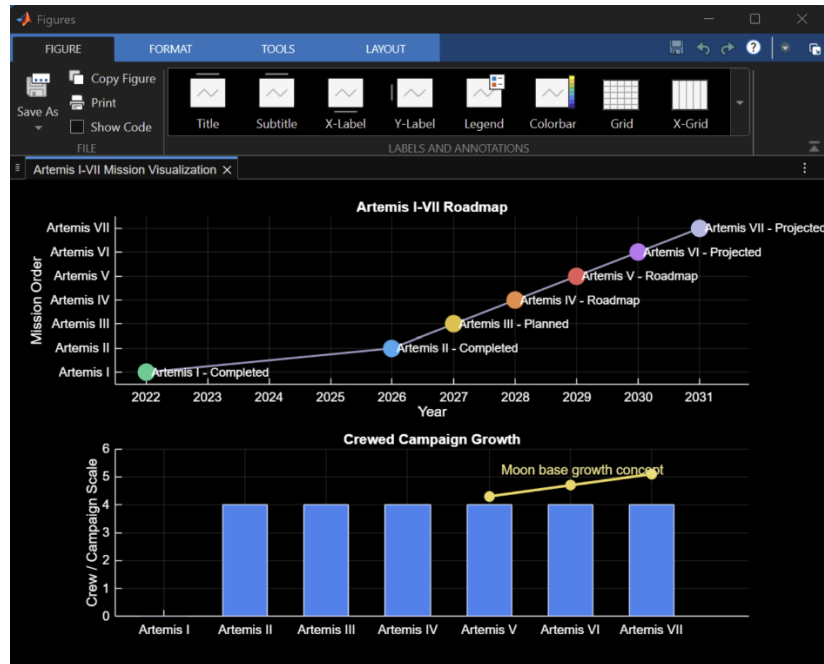


Figure (3): Mission Timeline Visualization Interface.

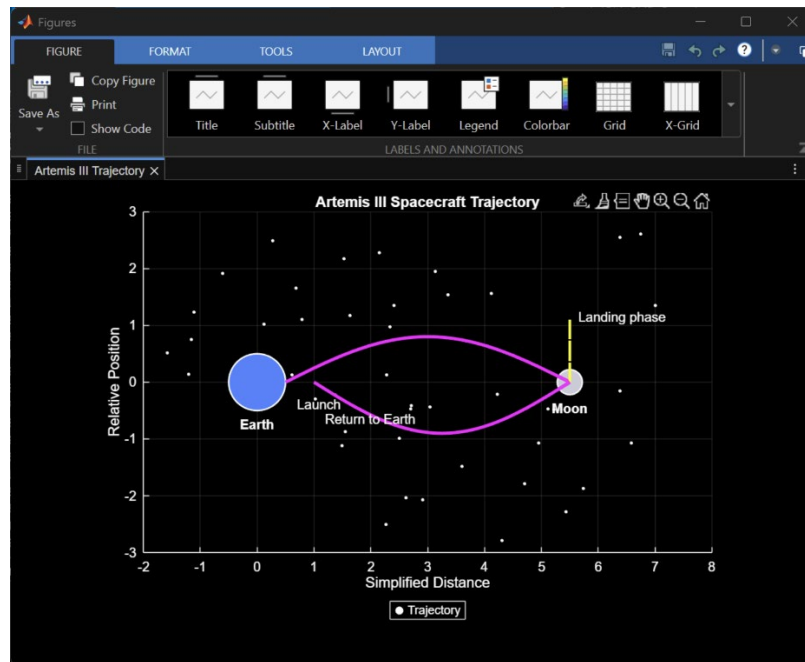


Figure (4): Trajectory Visualization Interface.